Analysis Report: Puma Sales Analysis Dashboard

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1. Project Overview

- **Project Name**
  Puma Sales Analysis Dashboard

- **Objective**
  To analyze Puma's sales performance across products, regions, and customer segments using transactional data and build a 5-page interactive Power BI dashboard. The dashboard uncovers revenue trends, profitability patterns, regional contribution, product performance, and customer behavior, enabling clear insights into data-driven decision-making.

- **Scope**
  The analysis focuses on:
  - Revenue and profit trends over time
  - Product level performance insights
  - Regional contribution to sales and orders
  - Customer segmentation based on gender and behavior

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2. Dataset Summary

Puma transactional sales dataset covering three fiscal years (2022–2024) across 11 Indian metro cities and multiple product lines.

- ~5000+ rows
- 14 columns

Key Characteristics

- Transaction level sales dataset
- Supports aggregation and filtering across multiple dimensions
- Clean structure for Power BI modeling

Key Entities Covered

- Orders
  - Contains order level details
  - Includes order date, order ID, and sales channel
  - Used to track total number of orders
- Products
  - Stores product related information
  - Includes product ID, product name, and product line
  - Contains pricing details such as cost and MRP
- Region
  - Contains geographical classification
  - Includes region name and region ID
  - Used for regional sales analysis
- Gender
  - Defines customer segments
  - Includes categories such as Men, Women, and Kids
  - Linked using gender ID
- Size
  - Represents product size information
  - Includes size ID and size labels like S, M, L, and Unknown
  - Helps with size-based sales analysis
- Sales (Fact Table)
  - Central table containing transaction data
  - Includes revenue, profit, units sold, and discount applied
  - Contains calculated metric profit margin
  - Connected to all dimension tables for analysis

3. Data Cleaning Steps Performed

- Removed or handled null values in:
  - Region
  - Size
  - Order_ID
- Standardized categorical fields
- Ensured numeric consistency in:
  - Revenue

    - Profit
    - Units Sold
- Handling Missing Values:
    - Critical fields → cleaned or removed
    - Noncritical fields → labeled as “Unknown”
- Validation
    - Revenue checked using Units Sold and MRP
    - Profit validated using cost and revenue

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4. Column-wise Assessment Summary

- **Order_ID**
    - Unique identifier for each transaction
    - Used to calculate total orders
    - No duplication expected
- **Order_Date**
    - Converted to proper date format
    - Used for monthly and yearly trend analysis
- **Sales_Channel**
    - Categorical column with In Store and Online
    - Used for channel comparison
- **Product_ID**
    - Foreign key linking to Products table
    - Ensures product level analysis
- **Product_Name**
    - Used for detailed product performance
    - Helps identify top selling items
- **Product_Line**

    - Category level grouping
    - Used in most visual breakdowns
- **Cost**
    - Base cost of product
    - Used for profit calculation validation
- **MRP**
    - Selling price reference
    - Used for pricing comparison
- **Region_ID**
    - Foreign key linking to region table
    - Enables regional analysis
- **Region**
    - Human Readable Region Name
    - Used in charts and filters
- **Gender_ID**
    - Links to Gender table
    - Enables segmentation
- **Gender_Category**
    - Men, Women, Kids classification
    - Used in customer analysis
- **Size_ID**
    - Links to Size table
    - Helps with size-based filtering
- **Size_Label**
    - S, M, L, Unknown
    - Used for product distribution insights

- **Revenue**
    - Core metric for sales performance
    - Aggregated across all visuals
- **Profit**
    - Key profitability metric
    - Validated cost and revenue
- **Units_Sold**
    - Quantity metric
    - Used for volume analysis
- **Discount_Applied**
    - Indicates discount impact
    - Help evaluate pricing strategy

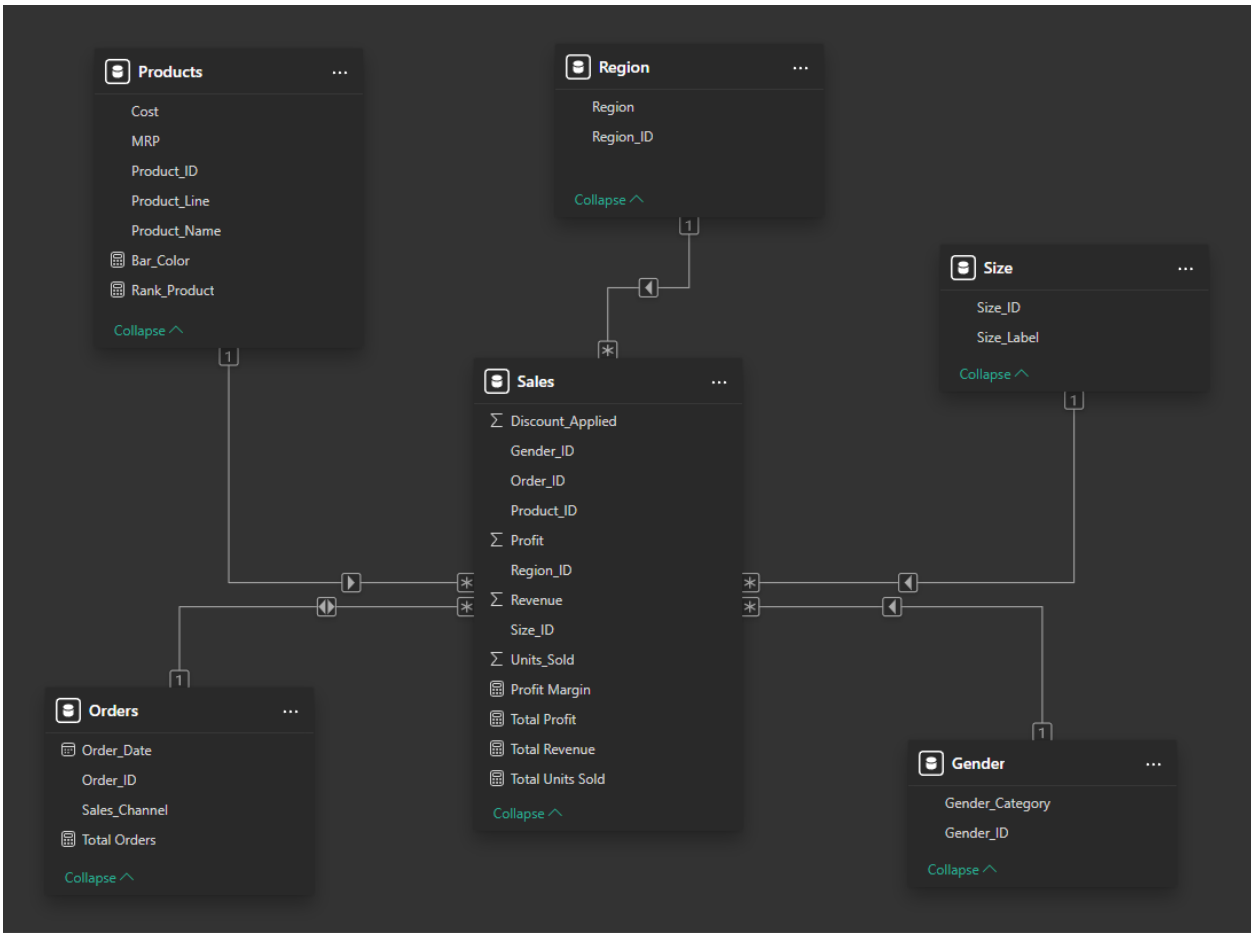
5. Data Model Overview

- Schema Type:

  The data was modeled using a Star Schema with a central fact table and supporting dimension tables to enable efficient analytical reporting.

Tables

- **Fact Table:**
  Sales – Contains core transactional metrics such as revenue, profit, units sold, discount applied, and profit margin. It acts as the central table for all analysis.
- **Dimension Tables:**
  Orders, Products, Region, Gender, Size – These tables provide descriptive attributes such as order details, product information, geographical data, customer segments, and size classification to support detailed analysis.
- **Relationships:**
  One to many relationships are established from each dimension table to the **Sales** fact table, enabling filtering and aggregation across multiple dimensions.



Key Calculations

Measures:

A set of DAX measures was created, including Total Revenue, Total Profit, Total Units Sold, Total Orders, and Profit Margin.
Additional measures were used to analyze revenue trends, product contribution, regional performance, and channel wise sales distribution.

Calculated Columns:

Derived columns were created for better segmentation such as cleaned Region labels, Size categories including Unknown handling, and structured Product Line groupings to support clearer analysis across visuals.

6. Analysis & Insights

- **Strong Revenue Performance:**
  The analysis shows consistently high revenue across regions, with total revenue reaching over 883M, indicating strong overall business performance.
- **Sales Trend Patterns:**
  Monthly analysis shows fluctuations, with peak performance around mid-year and noticeable dips in certain months, highlighting seasonal variation in sales.
- **Channel Performance:**
  Online sales contribute slightly more than In Store sales, indicating a growing preference for digital purchasing channels.
- **Product Performance Concentration:**
  A small group of top products contributes a significant portion of total revenue, showing dependency on key high performing items.
- **Category Level Insights:**
  Lifestyle and Basketball product lines generate higher revenue, while Training and Running show comparatively lower contribution.
- **Regional Distribution:**
  Cities like Pune, Ahmedabad, and Chennai lead in revenue generation, while "Unknown" regions contribute minimal value, indicating possible data quality issues.
- **Customer Segment Contribution:**
  Revenue distribution across Men, Women, and Kids is relatively balanced, with Men contributing slightly higher sales.

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7. Conclusions

The dashboard successfully transforms raw sales data into meaningful business insights across products, regions, and customer segments.

The analysis concludes that:

- **Revenue Stability:**
  The business maintains strong and consistent revenue across multiple dimensions.

- **Channel Shift:**
  Online sales are becoming slightly dominant, indicating a shift in customer buying behavior.
- **Product Dependency:**
  A few key products drive a large portion of revenue, creating both opportunity and risk.
- **Data Quality Impact:**
  Presence of Unknown categories affect accuracy and should be addressed.

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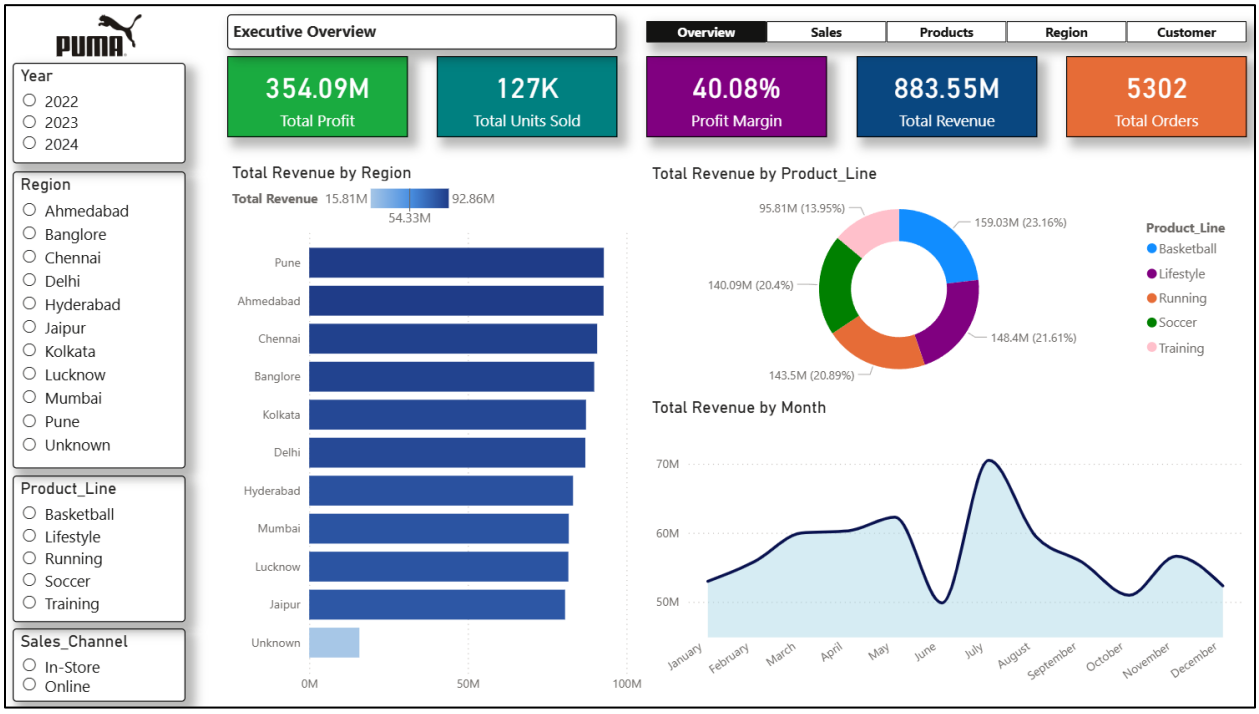
8. Recommendations

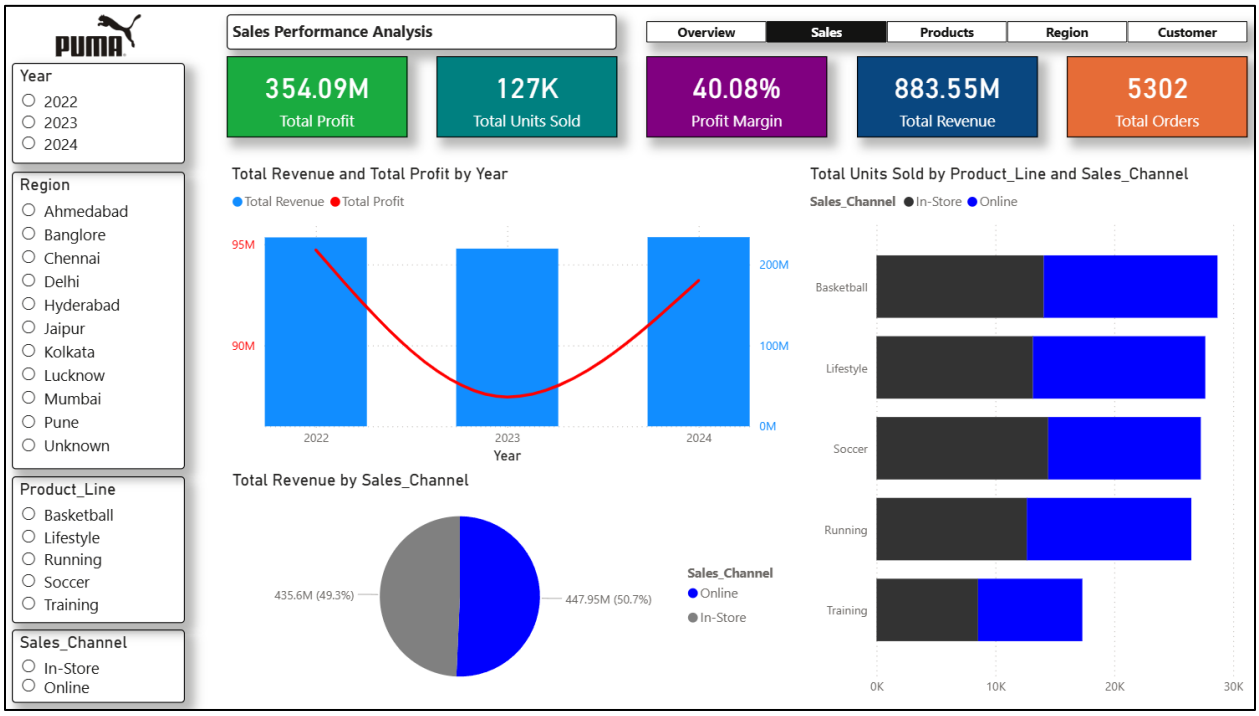
- **FOR SALES OPTIMIZATION:**
  Focus on promoting high performing products while improving visibility of low performing categories.
- **FOR PRICING STRATEGY:**
  Review cost and pricing for categories with lower profit margins to improve profitability.
- **FOR CHANNEL STRATEGY:**
  Strengthen online sales channels while maintaining in-store experience.
- **FOR DATA QUALITY:**
  Reduce Unknown values in Region and Size to improve analysis accuracy.
- **FOR REGIONAL GROWTH:**
  Invest more in high performing regions and analyze underperforming areas for expansion opportunities.
- **FOR CUSTOMER TARGETING:**
  Use gender-based insights to create targeted marketing and product strategies.

9. **Dashboards**

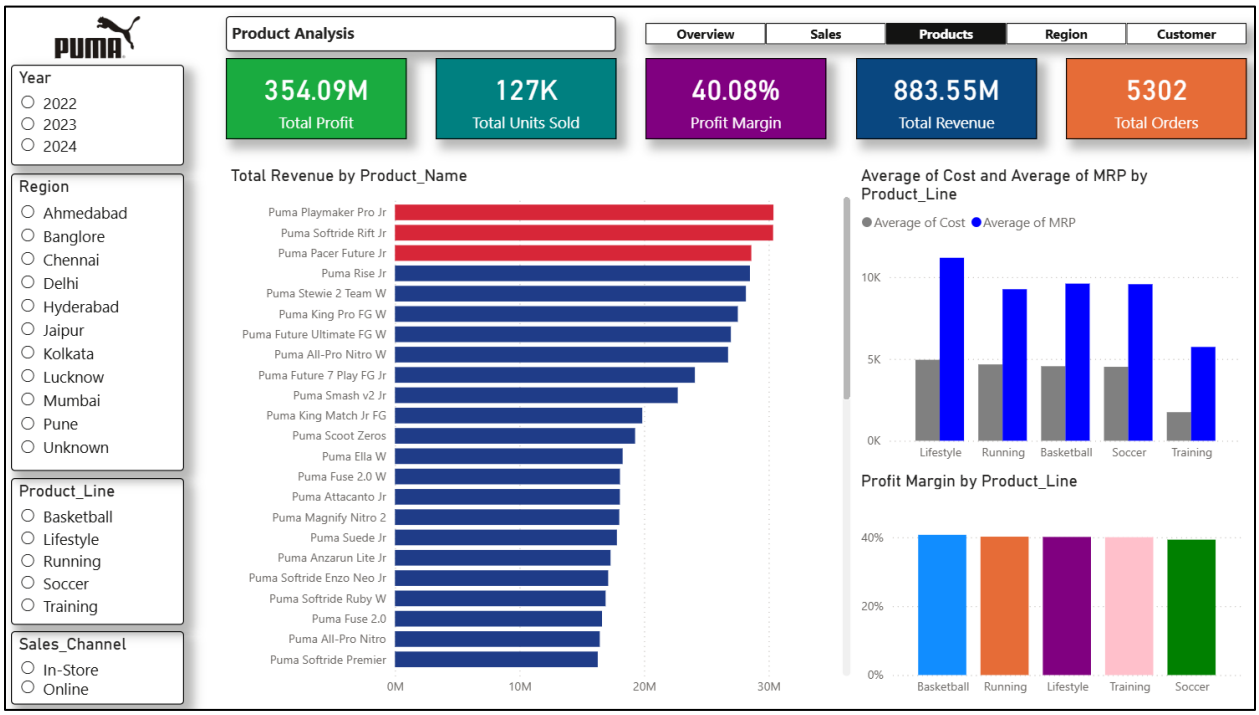
Executive Overview



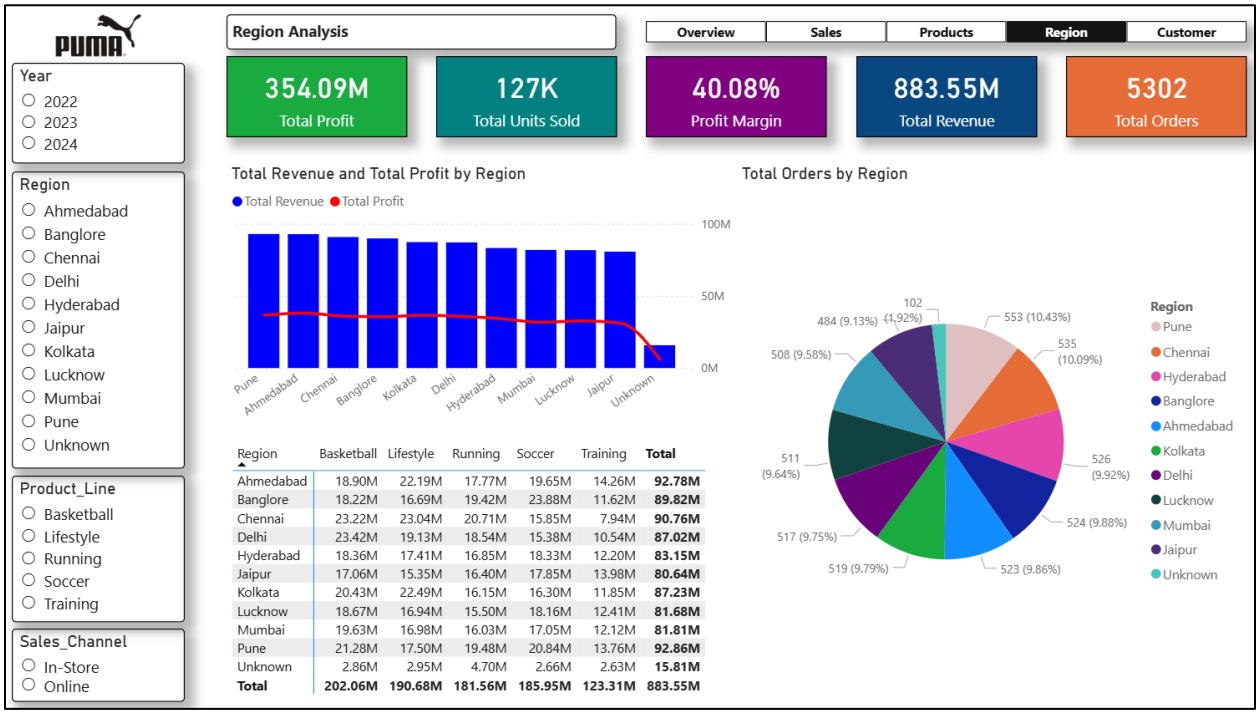
Sales Performance Analysis



Product Analysis

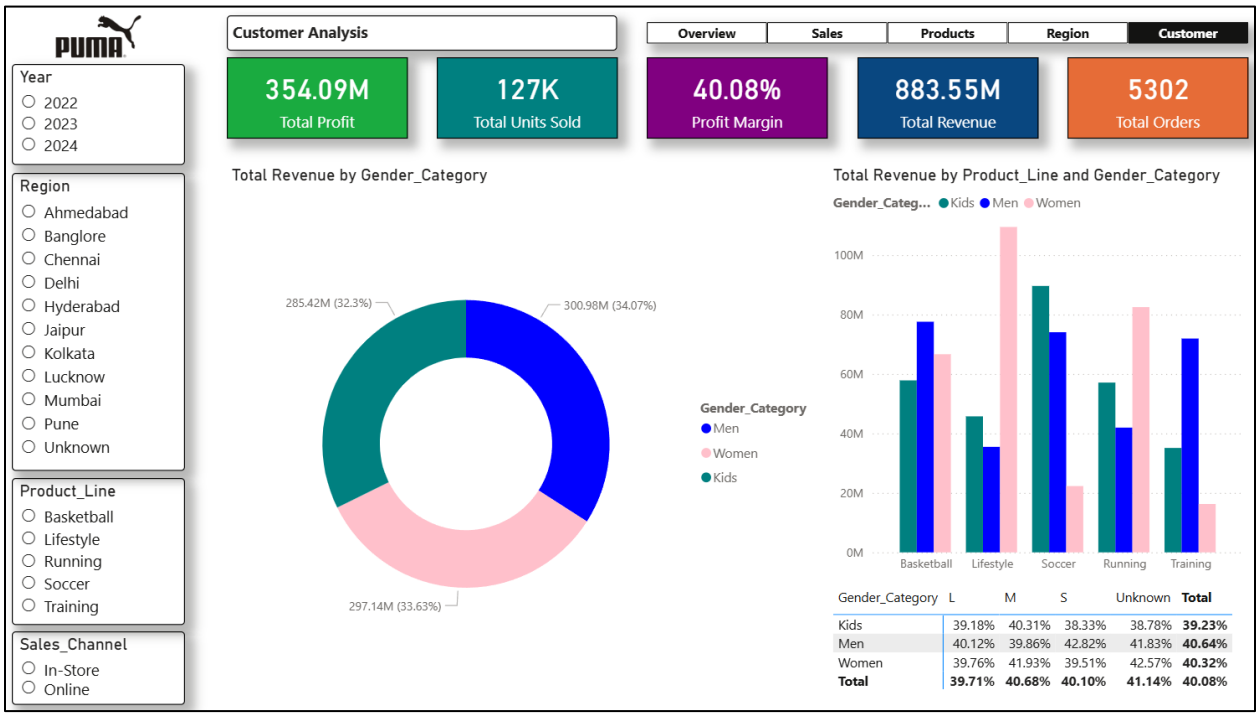


Region Analysis



Region	Basketball	Lifestyle	Running	Soccer	Training	Total
Ahmedabad	18.90M	22.19M	17.77M	19.65M	14.26M	**92.78M**
Banglore	18.22M	16.69M	19.42M	23.88M	11.62M	**89.82M**
Chennai	23.22M	23.04M	20.71M	15.85M	7.94M	**90.76M**
Delhi	23.42M	19.13M	18.54M	15.38M	10.54M	**87.02M**
Hyderabad	18.36M	17.41M	16.85M	18.33M	12.20M	**83.15M**
Jaipur	17.06M	15.35M	16.40M	17.85M	13.98M	**80.64M**
Kolkata	20.43M	22.49M	16.15M	16.30M	11.85M	**87.23M**
Lucknow	18.67M	16.94M	15.50M	18.16M	12.41M	**81.68M**
Mumbai	19.63M	16.98M	16.03M	17.05M	12.12M	**81.81M**
Pune	21.28M	17.50M	19.48M	20.84M	13.76M	**92.86M**
Unknown	2.86M	2.95M	4.70M	2.66M	2.63M	**15.81M**
Total	**202.06M**	**190.68M**	**181.56M**	**185.95M**	**123.31M**	**883.55M**

Customer Analysis



Gender_Category	L	M	S	Unknown	Total
Kids	39.18%	40.31%	38.33%	38.78%	**39.23%**
Men	40.12%	39.86%	42.82%	41.83%	**40.64%**
Women	39.76%	41.93%	39.51%	42.57%	**40.32%**
Total	**39.71%**	**40.68%**	**40.10%**	**41.14%**	**40.08%**